

UNIVERSITI KEBANGSAAN MALAYSIA
The National University of Malaysia

Making Quantum Dots and Nanotubes

KKKC6353 Nanoelectronic Devices — Lecture 2, Document 3 of 3

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How to use this document

Documents 1 and 2 explained why quantum wells, wires and dots behave as they do and what they are used for. This document is about how they are **made**. The choice of method matters: it decides how evenly sized the structures are, how perfect their crystals are, where they end up and how much they cost, and those in turn decide which devices they can go into.

The document follows the reading in the original L2 handout, *Fabrication processes of quantum dots*, and adds a section on carbon nanotubes. As before, each section opens with **Key ideas** and closes with **check questions**, and the optional green **Going further** boxes hold the details.

1 Two routes to the nanoscale

> **Key ideas**
>
> Top-down methods carve small structures out of a larger piece, like a sculptor.
>
> Bottom-up methods let atoms or molecules assemble themselves, like crystals growing.
>
> Top-down is best at putting things in exactly the right place; bottom-up is best at making them perfect and in huge numbers.

There are two ways to make something only a few nanometres across.

Top-down. Start with a larger piece of material and remove everything you do not want, as a sculptor carves a statue from a block of stone. This is how the chips of Lecture 1 are made. Its strength is **placement**: every structure goes exactly where it was drawn.

Bottom-up. Let atoms or molecules come together by themselves into the structure you want, as crystals of salt grow from a solution. Its strength is **quality and quantity**: each dot can be an almost perfect crystal, and billions are made at once.

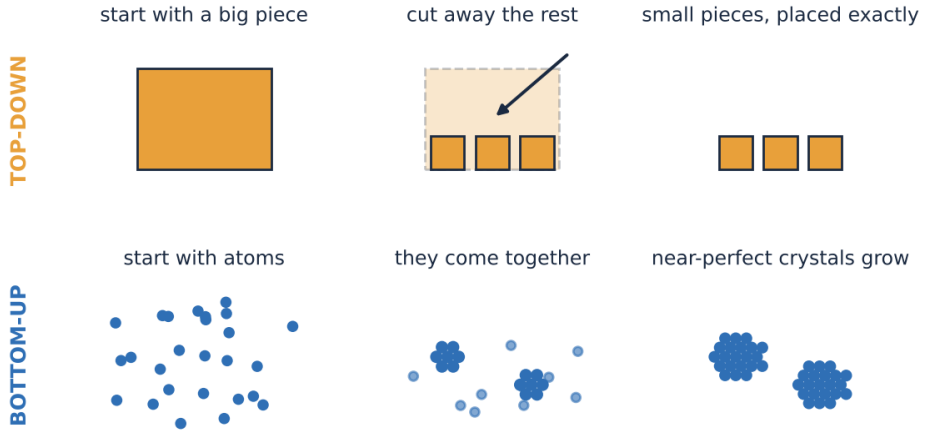


Figure 3.1. *The two routes. Top-down: start with a big piece and cut away the rest, leaving small pieces exactly where they were drawn. Bottom-up: start with atoms and let them come together into near-perfect crystals. Figure made for this course.*

The best modern devices often combine the two: bottom-up growth for the quantum structure itself, top-down processing for the device built around it.

Check your understanding

1. In one sentence each, describe the top-down and the bottom-up routes.
2. Which route is better at putting a dot in an exact place?

2 Top-down: carving

> **Key ideas**
> Lithography draws a pattern in a light- or electron-sensitive coating, and etching carves that pattern into the material underneath.
> An electron beam can draw features smaller than 10 nm, but only slowly, one point at a time.
> Carving leaves damaged edges, which spoil a dot's ability to give out light.

2.1 Drawing the pattern: lithography

Every top-down method begins by drawing a pattern. The wafer is coated with a thin film called a **resist**, which changes when light or electrons hit it. After the pattern is written, a developer washes away the unwanted parts of the resist. What remains protects the material beneath it while the rest is etched away.

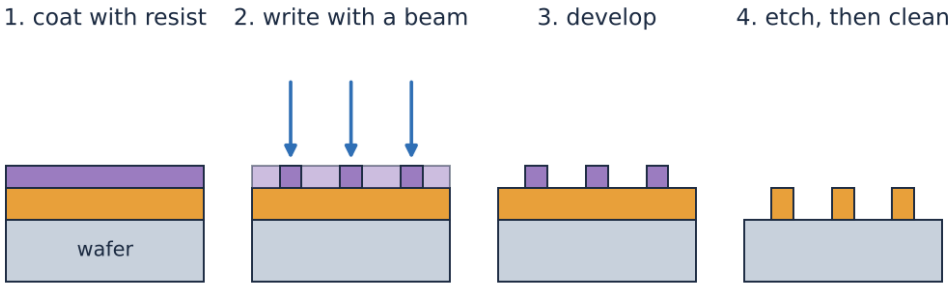


Figure 3.2. Lithography and etching in four steps: coat the wafer with resist, write the pattern, develop it, then etch away the unprotected material and clean off the resist. Small islands are left where the pattern was written. Figure made for this course; drawn schematically.

- **Electron-beam lithography.** A finely focused beam of electrons writes the pattern, like a very fine pen. It can draw features smaller than 10 nm, but it writes one point at a time, so a single wafer can take hours or days. It is used for research and for one-off devices, not mass production.
- **Photolithography.** Light shines through a stencil, called a mask, and prints the whole pattern at once. This is how chips are made in their billions (Lecture 1). Today's most advanced machines use extreme-ultraviolet light of wavelength 13.5 nm. Making the masks is very expensive, so photolithography pays off only when the same pattern is printed many times.

2.2 Carving it: etching

Etching removes the material that the resist does not protect. *Dry* etching uses a plasma, a glowing gas of charged particles, and cuts straight, steep walls. *Wet* etching uses a liquid chemical and is gentler but less precise.

Either way, the carved walls are rough and damaged at the scale of atoms. In a dot only 20 nm across, most of the atoms are close to that damaged surface. Electrons and holes that reach it are lost without giving out light, so etched dots glow poorly.

2.3 Other top-down tools

- **Focused ion beam.** A beam of heavy ions blasts material away directly, like a microscopic sandblaster. No resist is needed, which makes it handy for quick prototypes and for cutting cross-sections to look at, but the ions damage the crystal.
- **Dots made by voltage.** Instead of carving a dot, metal electrodes on the surface can push electrons out of the regions beneath them, leaving a small pool of electrons trapped in between. The walls are made by voltage, not by cutting, so there is no damaged edge, and the dot can be emptied until only a single electron is left. Dots like these are used as quantum bits in quantum computers, a topic for Lecture 3.

Check your understanding

1. What does the resist do in lithography?
2. Why do etched quantum dots give out less light than you might expect?

3 Bottom-up I: growing crystals layer by layer

> **Key ideas**
>
> Quantum wells are grown, not carved: machines lay down crystals one layer of atoms at a time.
>
> If a layer is squeezed to fit a slightly smaller crystal underneath, it breaks up by itself into tiny islands.
>
> These islands are self-assembled quantum dots, and they form without any pattern being drawn.

3.1 Building with atoms

The quantum wells of Document 2 are only a few nanometres thick, and their thickness must be right to within a single layer of atoms. They are made by **epitaxy**, growing a new crystal on top of an existing one so that the atoms carry on the same orderly arrangement. Two machines dominate:

- **Molecular-beam epitaxy (MBE).** Inside a chamber pumped to an extremely high vacuum, beams of atoms from heated ovens land on a heated wafer and build the crystal about one layer of atoms per second. Shutters can switch each beam on or off in a fraction of a second, so the boundary between two materials is sharp to a single layer. MBE is the method of choice for research and for the cleanest layers.
- **Metal-organic chemical vapour deposition (MOCVD).** Gases containing the atoms flow over hot wafers and react on their surface. It is faster and handles many wafers at once, so almost every LED and laser diode in the world is grown this way.

3.2 Dots that assemble themselves

Grow a material whose atoms are naturally a little further apart than those of the crystal underneath, for example indium arsenide (InAs) on gallium arsenide (GaAs), whose

spacings differ by about 7%. The first layer is squeezed to fit, and lies flat; it is called the **wetting layer**. But the squeeze stores up strain, like a carpet that is slightly too big for the room. After less than two layers of atoms (about 1.7 for InAs on GaAs), it becomes easier for the film to bunch up into small islands, where the atoms can spread out towards their natural spacing.

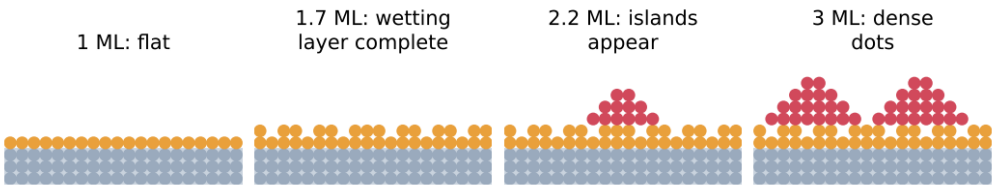


Figure 3.3. *Self-assembled dots, InAs (orange) growing on GaAs (grey). The first layers grow flat. Once the wetting layer is complete, at about 1.7 layers of atoms, extra material gathers into islands (red) that become quantum dots. Figure made for this course; drawn schematically.*

These islands, typically 20 nm wide and 5 nm tall, are **self-assembled quantum dots**. Covered with more GaAs, they end up buried inside a perfect crystal, with no damaged surface at all. This way of growing is named after Ivan Stranski and Lyubomir Krastanov, who predicted it in 1938, and is called **Stranski–Krastanov** growth. The dots are excellent light emitters and are used in lasers and in sources of single photons. Their drawbacks are that their sizes vary by about 10%, and that they appear at random places.

> **A common misconception**
>
> "Stranski–Krastanov growth is a hybrid of top-down and bottom-up." It is purely **bottom-up**: no pattern is drawn and nothing is removed, and the dots assemble themselves from the deposited atoms. Also, the dots appear after less than two layers of atoms, not after ten.

Check your understanding

1. What is the wetting layer?
2. Why does a strained film break up into islands?
3. Give one strength and one weakness of self-assembled dots.

4 Bottom-up II: chemistry in a flask

> **Key ideas**
>
> Most quantum dots in products are made by chemistry in a hot liquid, not on a wafer.
>
> A sudden burst of tiny seed crystals, all born at the same moment, then grows together, so the dots come out nearly the same size.
>
> The longer they grow, the bigger they get and the redder they glow.

4.1 Hot injection

The dots in quantum-dot televisions are made in a flask. The most famous recipe, the **hot-injection** method, was published by Christopher Murray, David Norris and Moungi Bawendi in 1993:

1. **Prepare** two solutions, one carrying the metal (for example cadmium or indium) and one carrying the other element (for example selenium).
2. **Inject** one quickly into the other, in a hot liquid at about 300 °C.
3. **Seed.** In a fraction of a second, a burst of tiny crystal seeds forms.
4. **Grow.** The temperature is lowered a little. The seeds grow, but no new ones form. The longer they grow, the larger they become.
5. **Stop** the growth by cooling, when the dots reach the size, and so the colour, you want.
6. **Clean** the dots by settling them out of the liquid and washing them.

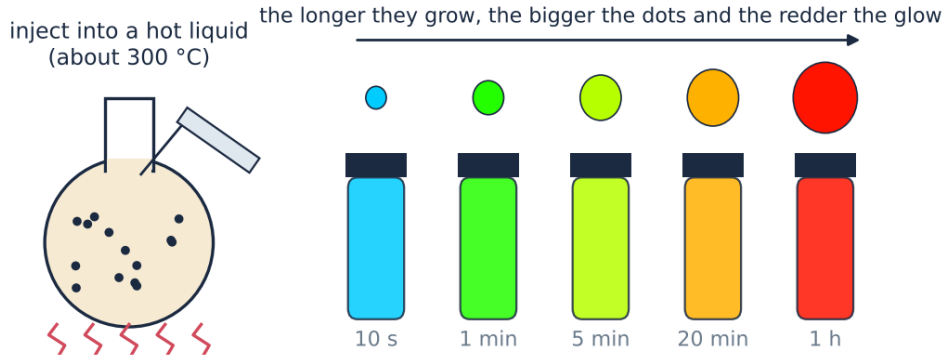


Figure 3.4. *Left: hot injection, in which one solution is injected quickly into another at about 300 °C. Right: samples taken out as the dots grow. Bigger dots glow redder. Figure made for this course; times are illustrative.*

The secret of the method is in steps 3 and 4. Because all the seeds are born at the same moment and then grow together, they end up almost the same size, often to within 5%. That evenness is what gives quantum dots their pure, narrow colours. Taking samples out at different times gives a whole rainbow of dots from one flask.

> **Going further: why the sizes come out even (optional)**
>
> Victor LaMer and Robert Dinegar explained this in 1950. Injection makes the concentration of dissolved building blocks shoot above the level needed to form new seeds. The burst of seeding uses them up so fast that the concentration falls back below that level, and no more seeds form. From then on the existing seeds only grow.
>
> While building blocks are plentiful, small dots grow faster than large ones, so the sizes even out. If growth goes on too long and the building blocks run low, small dots start to dissolve and feed the large ones, and the sizes spread out again. Growth is stopped before that happens.

4.2 A coat and some hair

A bare dot has broken bonds at its surface, which trap electrons and stop them giving out light. So the dot is usually given a **shell** of a different semiconductor, such as zinc sulfide (ZnS), around the light-emitting **core**. The shell keeps the electron and hole inside the core, away from the surface, and the share of absorbed light that is given out again rises from a few per cent to more than 90%.

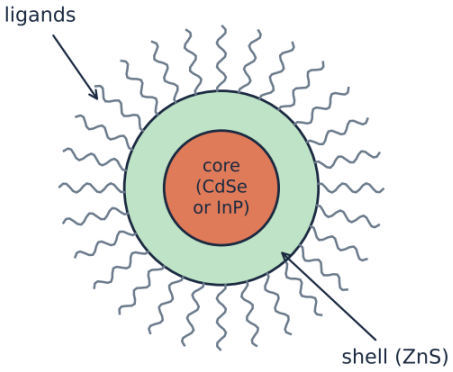


Figure 3.5. *A core–shell quantum dot. The core gives out the light, the shell protects it, and the ligands on the outside keep the dots apart in a liquid. Figure made for this course; drawn schematically.*

On the outside sit **ligands**, long molecules like hairs that stop the dots clumping together and let them be mixed into inks and films. For devices in which current has to flow from dot to dot, they are swapped for shorter ones.

4.3 Other recipes

The L2 handout lists several other ways of making dots from chemicals:

Method	How it works	Typical use
Heat-up	mix everything cold, then heat: simpler to scale up	display dots (CdSe, InP)
Room-temperature reprecipitation	dissolve the ingredients in one liquid, drop into another	perovskite dots
Hydrothermal	heat in a sealed, high-pressure vessel	oxide, sulfide and carbon dots
Microemulsion	tiny water droplets in oil act as size-limited reactors	ZnS, CdS and magnetic dots
Plasma	dots condense from a glowing gas	silicon and germanium dots
Flow reactors	mix and heat in narrow channels, continuously	large-scale production
“Green” synthesis	use plants, microbes or biomolecules	carbon dots

Perovskite dots, made since 2015 from a family of lead-containing crystals, glow very brightly and can change colour by changing their ingredients as well as their size. They are, however, easily damaged by moisture and heat.

Check your understanding

1. Why do dots made by hot injection all come out nearly the same size?
2. What does the shell around a dot’s core do?
3. A chemist wants red dots instead of green ones from the same recipe. What should she change?

5 Making carbon nanotubes

> **Key ideas**
>
> Nanotubes form when carbon atoms are freed at high temperature in the presence of tiny metal particles.
>
> Today most nanotubes are grown by chemical vapour deposition: a carbon-rich gas over metal catalyst particles in a furnace.
>
> Every method makes a mix of metallic and semiconducting tubes, so they must be sorted before they can be used in transistors.

5.1 Three ways to grow them

- **Arc discharge.** A large electric spark between two carbon rods vaporises carbon, which condenses into soot containing nanotubes. This is how Sumio Iijima found them in 1991. Adding metal to the rods produces single-walled tubes.
- **Laser ablation.** A powerful laser vaporises a carbon target mixed with a little metal in a hot furnace. Richard Smalley's group at Rice University used it in the mid-1990s to make high-quality single-walled tubes.
- **Chemical vapour deposition (CVD).** A carbon-rich gas, such as methane or ethanol, flows over tiny particles of a metal like iron or cobalt in a furnace at 600–1000 °C. The metal breaks the gas apart and a tube grows out of each particle. The tube's diameter follows the size of the particle. CVD is the most widely used method today, because it scales up and can grow tubes directly on a wafer, even lined up in neat rows on certain crystal surfaces.

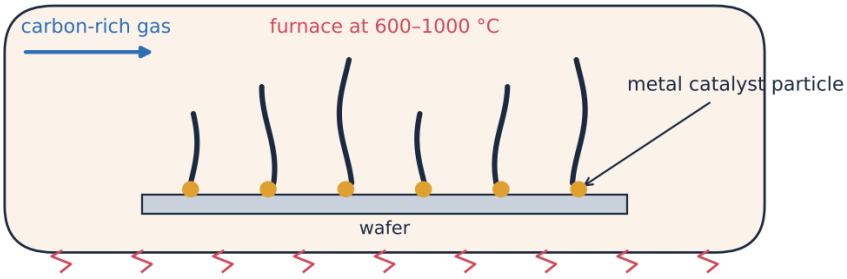


Figure 3.6. Growing nanotubes by chemical vapour deposition. A carbon-rich gas flows through a hot furnace over a wafer dotted with metal catalyst particles, and a nanotube grows out of each particle. Figure made for this course; drawn schematically.

5.2 Sorting and lining up

Document 2 explained that about one tube in three is metallic, and that a single metallic tube can short-circuit a transistor. No growth method yet makes only semiconducting tubes, so they are sorted afterwards:

- **Spinning.** In a very fast centrifuge, tubes coated with soap-like molecules settle at different depths according to their type and diameter.
- **Wrapping.** Certain long polymer molecules wrap only around semiconducting tubes and carry them into solution, leaving the metallic ones behind.

- **Burning off.** In a device, a large current can be sent through the tubes while the semiconducting ones are switched off, burning away the metallic ones.

Sorted tubes are then lined up on a wafer. In 2020 a group at Peking University made dense, aligned rows of tubes on 10 cm wafers that were more than 99.9999% semiconducting, close to what transistors need.

Check your understanding

1. Which method is most used to grow nanotubes today, and why?
2. Why do nanotubes have to be sorted before they are used in transistors?

6 Choosing a method

> **Key ideas**
> No method is best at everything: each trades placement, size control, crystal quality and cost against one another.
> The application decides which weaknesses can be tolerated.

The table compares the main methods, following the L2 handout.

Method	Size control	Crystal quality	Placement	Volume	Main uses
Electron-beam lithography and etching	good	poor at the edges	exact	very low	research devices
Dots made by voltage	set by the voltages	excellent	exact	low	quantum bits
MBE and MOCVD quantum wells	one atomic layer	excellent	whole wafer	high	lasers, LEDs, HEMTs
Self-assembled (Stranski–Krastanov) dots	moderate	excellent	random	high	lasers, single-photon sources
Colloidal (hot injection)	excellent	high, with a shell	none: dots float in a liquid	very high	displays, lighting, imaging
Nanotube CVD, then sorting	diameter set by the catalyst	high	improving	medium	nanotube transistors, research

Ratings are typical, not limits.

The pattern is the one Section 1 promised. Top-down methods put a structure exactly where it is wanted but tend to damage it. Bottom-up methods make near-perfect structures in huge numbers but cannot easily place them. A television needs billions of dots of the right colour but does not care where each one sits, so it uses chemistry. A quantum computer needs a few dots in exactly the right places, so it uses lithography and voltages. A laser needs perfect crystals in a thin layer, so it uses epitaxy.

Check your understanding

1. Which method would you choose to make red dots for a television, and why?

2. Which method would you choose for a single dot that must sit under a particular electrode?

Summary

1. **Top-down** methods carve structures out of a larger piece: lithography to draw, etching to cut. They place things exactly but leave damaged edges.
2. **Epitaxy** (MBE and MOCVD) grows crystals one layer of atoms at a time. Quantum wells are made this way.
3. **Self-assembled dots** form by themselves when a strained layer breaks up into islands after about 1.7 layers of atoms. This is a bottom-up process.
4. **Hot injection** makes colloidal dots in a flask. A burst of seeds that grow together gives dots of nearly equal size; growth time sets the colour; a shell makes them bright.
5. **Carbon nanotubes** are grown mainly by CVD from metal catalyst particles, and must be sorted to remove the metallic tubes.
6. **The application chooses the method**, according to which of placement, uniformity, quality and cost matter most.

Lecture 2 in one line. Small things are waves; trapped waves have only certain energies; the size of the trap sets those energies; and we now know how to build traps of the size we want.

Words to know

Word	Plain meaning
Top-down	Making small structures by carving a larger piece
Bottom-up	Making small structures by letting atoms or molecules assemble
Resist	A coating that changes where light or electrons hit it, used to draw patterns
Etching	Removing material that the resist does not protect
Epitaxy	Growing a crystal on top of another so that the atoms line up
Wetting layer	The thin flat layer that forms before self-assembled dots appear
Colloidal dots	Dots made in, and floating in, a liquid
Core–shell dot	A dot with a protective coat of a second semiconductor
Ligands	Molecules on the outside of a dot that keep the dots apart
CVD	Chemical vapour deposition: growing a material from a gas

Sources and further reading

Course material

- *Fabrication processes of quantum dots*, the original KKKC6353 Lecture 2 handout (in `Archived-Notes/`).

- G. Mishra, *Fabrication of quantum dots*, slides used as a source for this document (in `Archived-Notes/`).

Gentle introductions

- The Royal Swedish Academy of Sciences, *They added colour to nanotechnology*, popular science background to the Nobel Prize in Chemistry 2023 (nobelprize.org).

Original papers

- C. B. Murray, D. J. Norris and M. G. Bawendi, "Synthesis and characterization of nearly monodisperse CdE (E = S, Se, Te) semiconductor nanocrystallites", *Journal of the American Chemical Society* 115, 8706 (1993).
- V. K. LaMer and R. H. Dinegar, "Theory, production and mechanism of formation of monodispersed hydrosols", *Journal of the American Chemical Society* 72, 4847 (1950).
- D. Leonard, M. Krishnamurthy, C. M. Reaves, S. P. DenBaars and P. M. Petroff, "Direct formation of quantum-sized dots from uniform coherent islands of InGaAs on GaAs surfaces", *Applied Physics Letters* 63, 3203 (1993).
- S. Iijima, "Helical microtubules of graphitic carbon", *Nature* 354, 56 (1991).
- M. S. Arnold, A. A. Green, J. F. Hulvat, S. I. Stupp and M. C. Hersam, "Sorting carbon nanotubes by electronic structure using density differentiation", *Nature Nanotechnology* 1, 60 (2006).
- L. Liu et al., "Aligned, high-density semiconducting carbon nanotube arrays for high-performance electronics", *Science* 368, 850 (2020).

Figures. All figures were made for KKKC6353 (script in `figures-source/`).